

$$y' = x \cdot m + b \quad (\text{For data point } i: y'_i = m_1 x_{1i} + m_2 x_{2i} + b_i)$$

$$J(m, b) = \frac{1}{n} \cdot \sum (y'_i - y_i)^2 \quad \text{if } n=2 \text{ then}$$

$$J = \frac{1}{2} [(y'_1 - y_1)^2 + (y'_2 - y_2)^2]$$

$$\text{chain rule: } \frac{dJ}{dm} = \frac{dJ}{dy'} \times \frac{dy'}{dm} \quad \frac{dJ}{db} = \frac{dJ}{dy'} \times \frac{dy'}{db}$$

$$\frac{dJ}{dy'} = \frac{1}{n} \times 2(y'_i - y_i) = \frac{2}{n}(y'_i - y_i) \quad \text{let } e_i = (y'_i - y_i)$$

$$\therefore \boxed{\frac{dJ}{dy'} = \frac{2}{n} \cdot e_i}$$

$$\frac{dy'}{dm_1} = x_{1i} \quad , \quad \frac{dy'}{dm_2} = x_{2i} \quad \therefore \boxed{\frac{dy'}{dm} = x_i} \quad (\text{row } i \text{ of } x)$$

$$\frac{dJ}{dm} = \frac{dJ}{dy'} \times \frac{dy'}{dm} = \frac{2}{n} e_i \times x_i = \frac{2}{n} \sum_i e_i x_i$$

$$\therefore \boxed{\frac{dJ}{dm} = \frac{2}{n} x^T e}$$

$$y'_i = m_1 x_{1i} + m_2 x_{2i} + b_i$$

$$\frac{dy'_i}{db_i} = 1 \quad (\text{since } b_i \text{ was an added constant})$$

$$X^T = [(1, 4), (3, 10)]$$

$$X^T e: \text{row 1: } (1)(1) + (4)(11) = 1 + 44 = 45$$

$$\text{row 2: } (3)(1) + (10)(11) = 3 + 110 = 113$$

$$\frac{2 X^T e}{n} = \frac{dJ}{dm} = \frac{dJ}{dy'} \times \frac{dy'}{dm} = \frac{2}{2} \times [45, 113]$$

$$\frac{dJ}{dm} = [45, 113]$$

$$\frac{dJ}{db} = \frac{2}{n} e = \frac{2}{2} \times (1, 11) = [1, 11]$$

$$\frac{dJ}{db} = [1, 11]$$

Gradient descent rule:

new value = old value - (learning rate $\times$ gradient)

$$m\text{-new} = m - \alpha \frac{dJ}{dm}$$

$$= [-1, 2] - 0.01 [45, 113] = [-1 - 0.45, 2 - 1.13]$$

$$m\text{-new} = [-1.45, 0.87]$$

$$\frac{dJ}{db_i} = \frac{dJ}{dy_i'} \times \frac{dy_i'}{db_i} = \frac{2e_i}{n} \times 1 = \frac{2}{n} e_i$$

$$\therefore \frac{dJ}{db_i} = \frac{2}{n} e_i$$

Given $x = [(1,3), (4,10)]$

$$m = [-1, 2]$$

$$b = [1, 1]$$

$$y = [5, 6]$$

$$\alpha = 0.01 \text{ (used to minimize overshooting).}$$

$$x \cdot m: \text{row 1: } (1)(-1) + (3)(2) = -1 + 6 = 5$$

$$\text{row 2: } (4)(-1) + (10)(2) = -4 + 20 = 16$$

$$x \cdot m = [5, 16]$$

$$y' = x \cdot m + b = [5, 16] + [1, 1] = [6, 17]$$

Let error be e : $e = y' - y$

$$= [6 - 5, 17 - 6] = [1, 11]$$

$$J = \frac{1}{2} (y' - y)^2 = \frac{1}{2} (1^2 + 11^2) = 61$$

$$b_{\text{new}} = b - \alpha \frac{dJ}{db}$$

$$b_{\text{new}} = [1, 1] - (0.01) [1, 1] = [1 - 0.01, 1 - 0.01]$$

$$b_{\text{new}} = [0.99, 0.99]$$